# MANIPAL UNIVERSITY JAIPUR

Faculty of Engineering | School of Computer Science Engineering

Department of Artificial Intelligence & Machine Learning

Course Hand-out

**DEEP LEARNING LAB| AIM 3230 | 1 Credits | 0 0 2 1**

**Session:** Jan-May 2026 | **Program:** B. Tech (AI & ML) | **Semester:** VI

**Course Coordinator:** Ms. Akanksha Mrinali

## A. Introduction:

This laboratory course introduces undergraduate students to the practical foundations of Deep Learning through hands-on experiments. The course focuses on understanding and implementing neural network models using modern deep learning frameworks. Students begin with feedforward neural networks and optimization techniques such as gradient descent and backpropagation.

The course further explores Convolutional Neural Networks (CNNs) for image-based tasks, Recurrent Neural Networks (RNNs) including LSTM and GRU for sequential data, and introduces students to object detection, segmentation, and encoder–decoder architectures. Advanced topics such as autoencoders, Generative Adversarial Networks (GANs), and attention mechanisms are also covered at an introductory level to build strong conceptual and implementation skills suitable for B. Tech students.

## B. Course Outcomes

At the end of the course, students will be able to

| | |
|---|---|
| [AIM3230.1]: | Implement basic feed-forward neural networks using different activation functions and optimization techniques. |
| [ AIM3230.2]: | Apply gradient descent, backpropagation, and optimizers such as SGD, RMSprop, and Adam to train deep learning models. |
| [ AIM3230.3]: | Design and implement Convolutional Neural Networks (CNNs) for image classification tasks using standard architecture such as LeNet, AlexNet, and ResNet. |
| [ AIM3230.4]: | Implement Recurrent Neural Networks (RNNs), including LSTM and GRU, for handling sequential and time-series data. |
| [ AIM3230.5]: | Apply deep learning models to practical problems such as object detection, image segmentation, and generative modeling using autoencoders, GANs, and attention mechanisms. |

## C. Program Outcomes and Program Specific Outcomes

**PROGRAM OUTCOMES (PO)**

[PO.1]. Engineering knowledge: Apply the knowledge of basic science and fundamental computing in solving complex engineering problems

[PO.2]. Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

[PO.3]. Design/development of Computing solutions: Design solutions for complex IT engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the Information oriented public health and safety, and the cultural, societal, and environmental considerations

[PO.4]. Conduct investigations of complex problems: Use IT domain research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

[PO.5]. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

[PO.6]. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.

[PO.7]. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

[PO.8]. Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practices.

[PO.9]. Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse IT teams, and in multidisciplinary settings.

[PO.10]. Communication: Communicate effectively on complex computing engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

[PO.11]. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

[PO.12]. Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

**PROGRAM SPECIFIC OUTCOMES (PSOS)**

At the end of the B Tech AIML program, the student:

[PSO.1]. Will be able to design, develop and implement efficient software for a given real-life problem.

[PSO.2]. Will be able to apply knowledge of AI, Machine Learning and Data Mining in analysis big data for extracting useful information from it and for performing predictive analysis.

[PSO.3]. Will be able to design, manage and secure wired/ wireless computer networks for transfer and sharing of information.

## D. Assessment Plan:

| Criteria | Description | Maximum Marks |
|---|---|---|
| Internal Assessment (Summative) | Continuous Evaluation (Including lab Performance + Attendance (30)) + Quiz- 2 (10) | 40 |
| | Deep Learning project (Mid Term Evaluation (20)) | 20 |
| End Term Exam (Summative) | End Term Exam (20) + Viva (10) + Lab File (10) | 40 |
| | Total | 100 |
| Attendance (Formative) | A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking up the End Semester examination. The allowance of 25% includes all types of leaves including medical leaves. | |
| Homework/ Home Assignment/ Activity Assignment (Formative) | There are situations where a student may have to work in home, especially in mini-Project, these works are graded with marks. A student is expected to participate and perform the mini project and assignments with full zeal as participation by a student will be assessed and marks will be awarded. | |

## E. SYLLABUS

Introduction: Exploration of feedforward neural networks, including various activation functions and the application of gradient descent and backpropagation. implement different optimizers such as ADAM, RMSprop, and Stochastic Gradient Descent (SGD) to address optimization challenges. Convolutional Neural Networks (CNNs) by building and analyzing classic architectures like LeNet, AlexNet, and ResNet, while also applying regularization techniques like dropout and batch normalization. Recurrent Neural Networks (RNNs), specifically implementing LSTMs and GRUs to manage sequence data and address the vanishing/exploding gradient problem. semantic and instance segmentation, object localization using models like YOLO and RCNN, and a study of encoder decoder models. autoencoders, Generative Adversarial Networks (GANs), and the Attention Mechanism to understand their practical applications and internal workings.

## F. Textbooks & References:


1. Charu C. Aggarwal, Neural Networks and Deep Learning: A Textbook, Springer, 2023.

2. Vishnu Sharma, Somnath Dey, Practical Deep Learning for Cloud, Mobile, and Edge, 2nd Edition, O'Reilly Media, 2023.

3. Palash Goyal, Sumit Pandey, Karan Jain, Deep Learning for Natural Language Processing, 2nd Edition, Apress, 2023.

4. Satya Mallick & Tzutalin, Practical Computer Vision with Deep Learning, Packt Publishing, 2024.

5. Recent Research Articles (2022–2025) from IEEE TPAMI, CVPR, NeurIPS, ICLR, ACM Computing Surveys on CNNs, Transformers, GANs, and Attention mechanisms.


## G. Lecture Plan

| Lab No. | Topics / Experiments | Session Outcome | Mode of Delivery | Corresponding CO | Mode of Assessing the Outcome |
|---|---|---|---|---|---|
| 1 | Implement single layer feedforward neural network from scratch | Understand and implement basic feedforward neural networks | Experiment | AIM3230.1 | Continuous Evaluation |
| 2 | Implement gradient descent and backpropagation with different activation functions | Apply gradient descent and backpropagation for network training | Experiment | AIM3230.1 | Continuous Evaluation |
| 3 | Implementation of optimizers: SGD, RMSprop, Adam | Analyze and compare different optimization techniques | Experiment | AIM3230.2 | Continuous Evaluation |

| | | | | | |
|---|---|---|---|---|---|
| 4 | CNN architectures: LeNet, AlexNet, VGG, ResNet | Implement and analyze CNN architectures for image classification | Experiment Demonstration | AIM3230.3 | Continuous Evaluation |
| 5 | Regularization techniques: Dropout and Batch Normalization | Apply regularization techniques to improve model generalization | Experiment | AIM3230.3 | Continuous Evaluation |
| 6 | Recurrent Neural Networks: LSTM and GRU | Implement RNN-based models for sequential data analysis | Experiment Demonstration | AIM3230.4 | Continuous Evaluation |
| 7 | Object detection using YOLO and RCNN | Develop object detection models using deep learning techniques | Experiment Demonstration | AIM3230.5 | Continuous Evaluation |
| 8 | Semantic and Instance Segmentation | Perform semantic and instance segmentation on images | Experiment | AIM3230.5 | Continuous Evaluation |
| 9 | Autoencoders and Generative Adversarial Networks (GANs) | Understand and implement generative models | Experiment Demonstration | AIM3230.5 | Continuous Evaluation |
| 10 | Attention Mechanism and Encoder - Decoder Models / Mini Project | Apply attention-based models to solve practical problems | Experiment Demonstration | AIM3230.5 | Project Evaluation |

## H. Learner Centric Activities for Deep Learning Lab (AIM3230)

Learner-centric activities are designed to address diverse learning abilities and promote experiential, inclusive, and outcome-based learning. The activities are structured at multiple difficulty levels to ensure conceptual clarity for beginners while enabling advanced learners to develop higher-order skills.

**Low Difficulty Activities**

- Video-based learning: Short conceptual videos on neural networks, CNNs, and RNNs followed by basic MCQs and conceptual questions.
- Visual aids: Use of diagrams, flowcharts, architecture blocks, and training curves to explain backpropagation, CNN layers, and optimization.
- Hands-on guided experiments: Step-by-step lab exercises with starter code for implementing feedforward networks and CNNs.
- Worksheets: Worksheets for identifying activation functions, optimizer behavior, loss functions, and model components.

- Interactive activities: Think-pair-share discussions on model behavior, hyperparameters, and common training issues.
- Tool demonstrations: Demonstration of deep learning tools and libraries (TensorFlow / PyTorch) with observation-based tasks.

These activities support slow and beginner learners by providing guided, visual, and structured learning experiences.

**High Difficulty Activities**

- Advanced model implementation: Design and training of CNN, RNN, LSTM, and GRU models for real datasets.
- Model optimization & evaluation: Hyperparameter tuning, optimizer comparison, and performance analysis using accuracy, loss, and confusion matrices.
- Object detection & segmentation tasks: Implementation of YOLO / RCNN-based detection and semantic segmentation models.
- Generative modeling: Development of autoencoders and basic GAN models with result interpretation.
- Mini-project development: End-to-end development of deep learning applications such as image classification, object detection, or sequence prediction.
- Research exposure: Review and presentation of recent research papers or case studies in computer vision and deep learning applications.
- MOOC integration: Enrollment in relevant MOOCs (NPTEL / Coursera / edX) on Deep Learning or Computer Vision and application of learned concepts in lab tasks.

These activities enhance analysis, evaluation, problem-solving, and creation skills, preparing students for industry and higher studies.

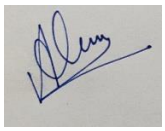
**Outside Class Engagement Activities**

| Type of Activity | Difficulty Level | Hours Engaged |
|---|---|---|
| Online Tutorial Viewing (DL basics, CNNs) | Low | 4 |
| Peer Learning / Group Discussion | Medium | 3 |
| Lab Practice & Debugging | Medium | 4 |
| Mini Project Planning & Design | Medium | 3 |
| Mini Project Implementation | High | 5 |
| Model Evaluation & Report Writing | Medium | 3 |
| Technical Webinar / Workshop | Low | 2 |
| Reflective Summary & Documentation | Low | 2 |

**I. Course Articulation Matrix**

CO–PO–PSO Mapping (1 = Low, 2 = Medium, 3 = High)

| CO | Statement | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **AIM3230.1** | Implement basic feed-forward neural networks and activation functions | 3 | 2 | – | – | 2 | – | – | – | – | 1 | – | 2 | 2 | 2 |
| **AIM3230.2** | Apply optimization techniques such as SGD, RMSprop, and Adam | 3 | 3 | 2 | 2 | 2 | – | – | – | – | 1 | – | 2 | 2 | 3 |
| **AIM3230.3** | Design CNN architectures for image-based tasks | 3 | 3 | 3 | 2 | 2 | – | – | – | – | 1 | – | 2 | 3 | 3 |
| **AIM3230.4** | Implement RNN, LSTM, and GRU for sequence modeling | 3 | 2 | 2 | 2 | 2 | – | – | – | – | 1 | – | 2 | 2 | 2 |
| **AIM3230.5** | Apply deep learning models for detection, segmentation, and generative tasks | 3 | 3 | 3 | 3 | 2 | 1 | – | – | 1 | 2 | – | 3 | 3 | 3 |



**Course Coordinator**

**Name:** Ms. Akanksha Mrinali

**MUJ ID.** MUJ2008

**Head of the Department**

**Name:** Dr. Deepak Panwar

**MUJ ID.** MUJ1158

**Student Representative**

**Name:**

**Registration No.**